Data Mining 1 (2026) 1–24 Submitted 12/06; Published 12/15

Counter-Strike 2 Round Outcome Prediction

Augustas Markevičius[1] AUGUSTAS.MARKEVICIUS@MIF.VU.LT
Petras Radžiūnas[2] PETRAS.RADZIUNAS@MIF.STUD.VU.LT
Titas Burinskas[2] TITAS.BURINSKAS@MIF.STUD.VU.LT
Tomas Giedraitis[1] TOMAS.GIEDRAITIS@MIF.STUD.VU.LT
Data Science[1] *and Bioinformatics*[2] *study programmes*
Faculty of Mathematics and Informatics[1]*, Life Sciences Center*[2]
All authors contributed equally

Advisor: Jurgita Markevičiūtė

Abstract

The Counter Strike 2 Round Outcome Prediction (CS2-ROP) project aims to predict the outcome of rounds played in the video game Counter Strike 2 using in-game metrics, with a primary focus on player location data in the game map. Machine learning models are used that take the selected in-game metrics at snapshot time frames during a round as input, and produce a quasi-real time prediction of which team is more probable to win the round.

Keywords: Outcome prediction, Counter-Strike, Machine Learning

1. Introduction

Note: This introduction is unfinished and unpolished. It is included only to help the reader understand the subject matter.

Our data mining research investigates the game of Counter-Strike 2 (CS2) by trying to predict the outcome of the round based on a player's position on the map and other factors like side, health, weapons, team economy, etc. In a CS2 game, a single map is played until one of the teams wins 13 rounds out of 24. In the case of a tie, 12-12, teams play 6 tie-breaker rounds until one team wins 4 rounds. There are two sides in each match, Counter-Terrorists (CT) and Terrorists (T). Each map is played differently due to different geometry that produces unique game situations, timings, and optimal positions for each side. Due to the unique nature of each map, in order to predict a round-win outcome, all factors must be taken into consideration on a map-to-map basis.

The core premise of why researching CS2 is particularly interesting is because the players in the game operate on limited knowledge under certain fog-of-war and intrinsically perform probabilistic estimations about the intentions and the positions of the enemy side. While dueling is a completely mechanical skill, and may take a second to end, estimations about enemy positions and optimal ways to counter them are purely informational and take up the majority of the game. So probabilistic models might shine light not only about some extrinsic qualities of the game, like what factors determine win outcomes the most, but also give insights into how the game is played in itself. Other games also have similar qualities like fog-of-war or prediction of the opposing side's intentions, however, CS2 is unique in the fact that due to its digital nature, every event and facet of the game is stored as information

©2026 Augustas Markevičius, Petras Radžiūnas, Titas Burinskas, and Tomas Giedraitis.

and can be extracted as data and reproduced near-verbatim, making it a desirable medium for data mining.

Round win conditions for each side are different. For Ts it is either:

1. Planting the bomb at Bombsites and successfully running out the timer (40 s) of the bomb
2. Eliminating all CTs.

For CTs it's:

1. Defusing the planted bomb,
2. Eliminating all Ts,
3. Running out the round timer (115 s), as long as a bomb is not planted.

Due to this dynamic, Ts are considered the attacking side, and CTs the defending side. Round win-rates are fairly even for both sides on most maps played at the professional level, though with a slight skew towards the CT side. Still, some maps retain equal side win-distribution, such as the quintessential "well balanced" map Dust2.

The reason why the aforementioned parameters were chosen is because all of them are expected to produce a greater consistency in data, while also removing uninformative randomness. The null hypothesis of this research is that round outcomes are random and don't depend on player positions and other in-game factors, therefore it is required to remove out-of-game random factors as much as possible in order not to conflict the two. As with all games, there is an element of randomness, which must be controlled for in order for a game to be considered competitive (i.e. skill-based) and predictable. It is a somewhat desirable quality for a game or a sport to be predictable "on-paper", where quantitative features inform the outcome of the game, and statistics are often used for this reason.

In CS2, professional player performance is generally more consistent than that of casual players, because of the regularity of the team roster, effective team communication, knowledge of the optimal strategy in different scenarios and high individual skill. The teams also have greater parity with each other and consistently outperform teams outside of top 10. The teams, no matter the period, are a benchmark of 'how the game should be played' at the highest skill level.

The same can be said for matches played in LAN tournaments as opposed to On-Line games, where match outcomes are more predictable due to better connectivity quality, uniformity of environment (that is, all players play in one area, rather than in separate locations) and other factors which in on-line games are harder to define quantitatively.

For our research, we chose to investigate professional games of the top 10 teams playing each other based on hltv.org CS Rating 3.0 (teams) which is, if not one of the most robust rating systems for CS2 player performance ranking, at least the most widely accepted and used. The hltv.org database evaluates whether the teams at the time of their match were in the top 10 and filters out only these results. Of all types of matches, only LAN (Local Area Network) tournament matches were selected. The time frame chosen spanned from 2026-08-01 to 2026-10-11, totaling 72 days, during which 60 matches were played. These

matches consisted of games played in a "best-of" format, and for each game a different map is picked. The majority of matches were BO3 (Best of 3), with BO5 occurring rarely, and BO1 only once. 150 games overall were played. We've used convenient in-built data filtering options in hltv.org/stats for queries and slicing. We chose the two most played maps, Dust2 and Mirage. Of the 150 games played in the 60 matches of the period, 30 were on Dust2 and 30 on Mirage, together making up 40% of all games. At least one of the two maps was played in 47 of the 60 matches (78.3%).

Other than their popularity, there were other reasons that these two maps were picked. For the map Dust2, the CT/T round win-rate distribution is 49.7%/50.3% respectively, almost an ideal 50-50 "coin-toss", so to say. Mirage has greater variance, with 55.8%/44.2% distribution. We find this difference an interesting point of investigation, by trying to see how our model will fare for maps which have side-oriented and side-independent round outcomes. These two maps are well-known in pro-play and have been played for decades, so win-rate distribution isn't a sort of recency-bias, but a well set-in "quality" about the map itself. While analyzing all maps would have been ideal, only two maps were picked to reduce both the amount of data produced and the computational requirements to analyze the data. We assume that this leaner approach will be enough to investigate the differences between the two maps.

The short time frame of 72 days was chosen due to the ever-changing nature of online competitive video games, where updates can alter the game drastically. Nevertheless, some changes within the game, team rosters, tactics, etc. do occur over time and cannot be avoided completely. All of these decisions can be backed more robustly with various separate analyses, but this falls outside the scope of our work due to time constraints.

In general, many of the decisions were also driven by the fact that our approach will require machine learning model training. In an initial data test, a single match played on one map produced 1,600,000 rows of unfiltered tabular info. It is assumed that the amount of possible interactions between variables will be extremely high, thus narrowing will make it more manageable for the intended scope of our research.

One of the key challenges in initializing a relevant database is parsing the Demo files that store the entire information about a game tick-by-tick. Tick rate is the rate of recorded instances per second. Actual matches have 128 tick rate, CS2 Demo files have 64 tick rate to save space. While this may incur some loss of information, 64 instances per second is considered enough to account for all of the important events, except micromovements or minuscule information about crosshair placement or accurate hitbox hit instances.

2. Literature Review

Machine learning methods have been applied for sports outcome prediction, for example football (Moya et al. (2025), Berrar et al. (2024), Mills et al. (2024)), basketball (Chen et al. (2021), Ouyang et al. (2024), Alves and Barbosa (2025)), and esports (Stanlly et al. (2022), Makarov et al. (2017), Shen (2022)). However, the niche of research concerning CS2 (and by extension Counter Strike: Global Offensive, the game version before CS2) is very narrow, especially with regards to match or round outcome prediction. For that reason, this literature review mainly seeks to highlight the studies which have the most relevance

to our established goal, that is round outcome prediction using in-game metrics, as well as reviewing publicly available tools that have sought to achieve similar goals.

Of primary interest was the work done by Xia et al. (2025) To the best of our knowledge, it is the only study focused on round win and match win prediction using spatial positioning as one of the key predictors. XGBoost, Logistic regression and Decision tree ML methods were implemented. While the data was limited to CS:GO professional matches in 2022 and only the Dust2 map, the results showed that both XGBoost and Logistic regression could be used to effectively predict match wins with approx. 86% accuracy, and player death in a given time window with approx. 66% accuracy, with key predictors being economic factors, spatial positioning and team coordination.

Less relevant, but still noteworthy are two studies by Jing et al. In Jing et al. (2026a), the authors used machine learning methods (primarily XGBoost), along with SHAP for intrepretation, to distinguish key performance metrics in CSGO and CS2 professional players that correlate with a higher HLTV player rating. While this study is very valuable in determining performance characteristics that could possibly be used as features for predictions, it is also limited by the fact that the features used were highly correlated, and that the response variable is a HLTV rating which has been turned into a binary value representing "good" and "okay" players. In Jing et al. (2026b), the authors continued their work in identifying key performance metrics by conducting network analysis of how different metrics associate with each other. While the strongest associations found were rather obvious, e.g. the metric "Kills per round" being highly associated with "Rounds with kills", an interesting finding was that the dominant network nodes shifted from an aggression focused play-style in CS:GO, to a more balanced play-style prioritizing a mixture of defense and offense in CS2. This could mean that features selected for CS:GO match or round prediction may not be suitable for CS2, and vice versa.

Eshrah and Hassan (2025) investigated a machine learning approach to real time round outcome prediction, taking into account side, round number, economy management, equipment, win streak and player performance metrics. The methods used were XGBoost, Long short-term memory neural network and Random Forest. The models they trained showed very impressive prediction accuracies of around 95%. However, a few glaring issues are noted, such as no mention of whether data used was of professional or casual play, whether the current round was predicted using only previous round data or together with real time current round data, and no mention of tests for model overfitting. Still, the work is of interest because it specifically targets CS2 and takes into account win streak data, under the assumption that this could reflect psychological stressors, for example a loss streak leading to players under-performing due to added stress.

Finally, while not directly relevant to our goal, Björklund et al. (2018) used a neural network approach to predict CS:GO match outcomes. The interesting part of the work is that rather than using player data for direct prediction, instead the data was clustered into groups which were intended to represent unique play styles. These groups would then be used to evaluate the composition of opposing teams and that evaluation used to predict the match outcome. While the accuracy of the model was modest, being around 61 - 65%, conceptually this work is of interest because it demonstrates that team composition based on play-style can be used as a metric for match outcome prediction.

Moving on to the publicly available tools. One of the most similar in goals and structure is the tool cs2-win-prediction (TaiZo1 (n.d.)), a data pipeline, exploratory analysis and classical modeling project which compares team statistics of top teams (based on hltv.org statistics) across multiple maps. Snapshots are taken at freeze_end + 2s (after players purchase equipment, but before any duel). Round outcome prediction is based on three logistic regression models, one each for pistol rounds, post-pistol rounds and normal rounds. Three takeaways are noted by the authors, that have potential for testing with our own research: "1. The round type is the strongest predictor of predictability, pistol rounds stay barely above chance; post-pistol rounds are almost deterministic; 2. The immediate round state is the dominant signal. Adding HLTV stats or map identity gains at most +0.01 AUC over the round-state baseline on normal and post-pistol rounds; 3. PCA does not yield a useful dimensionality reduction. 12 components (out of 28 to 43 input features) are needed to capture 80 % of inertia in every regime."

A recently updated project csmapwinpredict (wyn0n0 (n.d.)) has a very similar scope of analyzing Demo file for predicting the current round winner of CS2 matches in real time (updated approximately once per second). One of its key advantages is the live demo replay function, where one can see win-rate probability as the replay goes, a potentially useful feature for testing. While not stated explicitly, a look at the round win probability evaluation code shows the use of sklearn's DummyClassifier, ColumnTransformer, SimpleImputer and LogisticRegression modules, indicating the project to be based on machine and deep learning methods. However, there are limitations, as stated by the project authors: "Predictions and map settings currently focus on Mirage. Results on other maps have not been verified. Large Demos may require substantial processing time and memory." The fact that the tool is based on Mirage serves our research, though only partially and some reservations have to made. Additionally, the slow processing time of large demos could become a liability if scaling a solution would ever be considered, and our project prefers to have that option in case we need a greater dataset.

Another CS2 prediction tool is CS2_RoundPrediction (ratulsarker (n.d.)) which "Predicts which team (CT or T) will win a given round in CS2 using deep learning on mid-round game state snapshots." The MLP Tuned (Optuna, 50 trials) model showed the best results, MLP Standard (86 features) coming second and KNN (k=3) coming in third. Other methods used were Logistic regression, Random Forest, LSTM and XGBoost. Top predictive features were 1. man_advantage (player count differential, 29.8%), 2. equipment_advantage (8.4%), 3. health_advantage (4.7%), 4. and t_alive, kills_this_round_t, armor/helmet_counts (presumably < 4.7 %). Their analysis involved 530 matches, 9,746 rounds and 161,271 snapshots, with 86 features extracted from mid-round game state, and a class balance of 48.9% CT wins / 51.1% T wins.

Another lean but effective round outcome predictor project is CSGOPredictor (d-roho (n.d.)). This tool used Logistic Regression with a robust, verified and prepared dataset of 700 demos for model training to predict CS:GO round outcomes. The total number of snapshots in the dataset were 122411, with the snapshots being i.i.d, treated as individual data points. The project also provides data analysis documentation, which other sources lack. It is one of the more used datasets for many similar projects and comparisons of results and methods can readily be made between them.

Two deep learning based tools for CS2 are worth mentioning here, despite deep learning falling outside of the scope of our work. cs2-evalbar (eigenpaul (n.d.)) is a neural network based tool, using a "roughly 70K-parameter GraphTemporalTransformer over a 20-second, 4 Hz state window." It claims meaningful results by game phase (<15s, 30-60s >=90s) with accuracies of 0.60, 0.68, 0.88 respectively. Another interesting finding is that the project claims "extra capacity overfits near 95 groups, but widening can help once the dataset reaches 142 groups." One big caveat is that the application was trained on Inferno, a map which we will not use in our work, so the results could be difficult to compare.

cs-net (Gary2005 (n.d.)) is a transformer-based path prediction & game-state forecasting tool for Counter-Strike 2 trained with Pro architecture (d_model=768, 138.7M params). While the intended use is for player movement, per-player win rate, survival probability and future kill probability prediction, it is still an interesting example for comparison between DL and ML solutions. However, the detailed documentation is in Chinese, so correct reproducibility may be unreliable because of translation.

3. Data

3.1 Data set

From the hltv.org database (hltv), the top 10 teams by hltv ranking were chosen and 60 matches were downloaded as demo files for the maps Dust2 and Mirage in which these teams were playing. Awpy was used to parse the data in the demo files. A total of 1257 rounds were parsed, generating various categories of parsed data, such as economy, kill and death, round state data, etc.

The names of the top 10 teams selected were: 9z, Aurora, FURIA, FUT, FaZe, Falcons, G2, Legacy, MOUZ, NAVI (also referred to as Natus Vincere), Spirit, Vitality

For exploratory data analysis purposes, round state data (round_states) was created by aggregating round, bomb, equipment, economy and health data team-wide for each 1 s snapshot during a round. This was combined with non-snapshot round data (rounds) to produce an enriched round state dataset (round_states_enriched). Information about these three datasets is presented in Table 1. The variables contained within the enriched round data dataset are presented in Appendix A.

Table 1: Information about the round, round state and enriched round state datasets

Dataset	Rows	Columns	Missing values	Matches	Rounds
rounds	1257	26	1288	60	1257
round_states	116201	45	195710	60	1257
round_states_enriched	116201	59	250917	60	1257

3.2 Exploratory data analysis

Using the data in the enriched round states dataset, exploratory data analysis was performed. First we present some information about the dataset itself.

Table 2 lists the missing values in the enriched round dataset. seconds_since_plant, plant_site and plant_tick have many missing values because they are computed only if a

bomb plant occurs during a round. ct_won_previous_round can only be computed from the second round onwards, thus values are missing for the first round. The remaining missing values belong to the match Falcons vs Natus Vincere on the Mirage map, due to an error during demo file parsing. Data imputation was not attempted. If imputing the missing data for this match will prove to be impossible, then it will be removed from the dataset.

Table 2: Missing variables in the enriched round data dataset

Variable	Missing values	% Missing
seconds_since_plant	97855	84.2
plant_site	97855	84.2
plant_tick	47215	40.6
ct_won_previous_round	4116	3.5
reason_name	1263	1.1
ct_buy_type	1198	1.0
t_buy_type	1198	1.0
first_death_side	217	0.2

Table 3 lists the number of matches and rounds played for each map.

Table 3: Number of rounds and matches played by map

Map name	Matches played	% Rounds played
Dust2	30	628
Mirage	30	629

Next, we provide some data related to in-game metrics. Figure 1 provides information about round duration by map. The distribution looks approximately normal, with both the Dust2 and Mirage maximum appearing at about 115 seconds, which is how much time a single round can take without a bomb plant occuring.

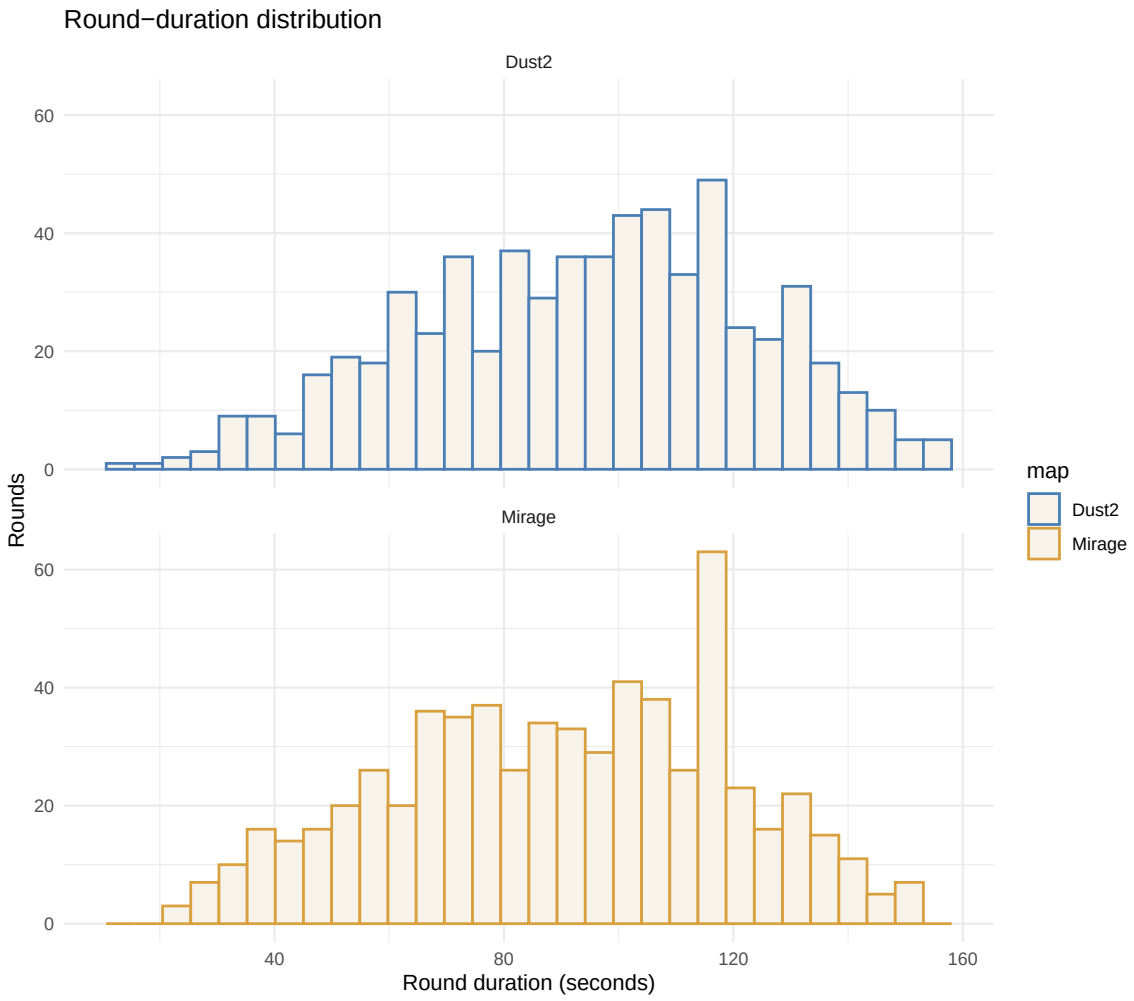


Figure 1: Round duration distribution by map type

Figure 2 provides information about round-win share by map and side. Dust2 seems to have a very balanced win share for both sides, while Mirage favors the CT side.

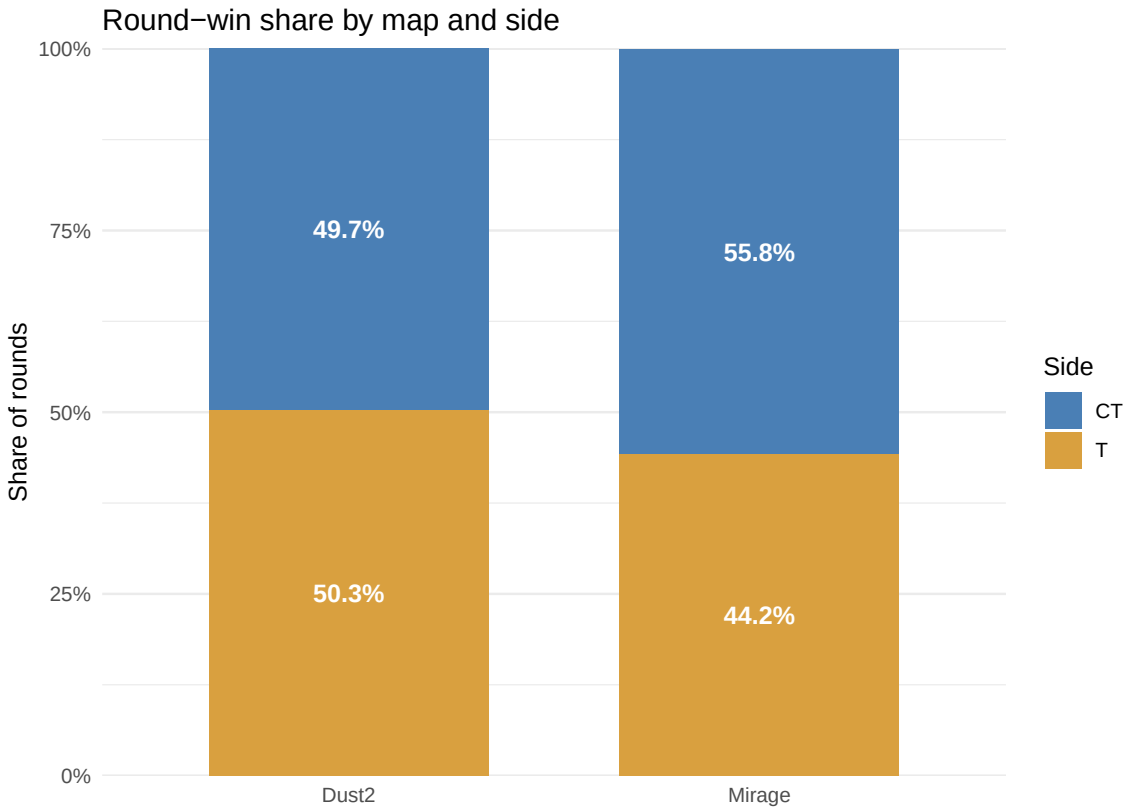


Figure 2: Side win share by map

Table 4 and Figure 3 shows the win types by map. The missing values the Mirage map are from the previously mentioned match with error in data parsing. ct_win and terrorist_win types are achieved by one side killing all players from the other side without a bomb being planted. target_bombed is a T side victory by bomb plant with the CT side not being able to defuse it in time, and bomb_defused is the CT victory type where they manage to defuse the bomb in time. target_saved is a CT side win by running out the timer, that is the T side is unable to plant the bomb in time.

Table 4: Win types by map

Map	Win type	Rounds with this win type	% Rounds
Dust2	bomb_defused	79	12.6
Dust2	ct_win	210	33.4
Dust2	target_bombed	129	20.5
Dust2	target_saved	23	3.7
Dust2	terrorists_win	187	29.8
Mirage	bomb_defused	75	11.9
Mirage	ct_win	225	35.8
Mirage	target_bombed	105	16.7
Mirage	target_saved	43	6.8
Mirage	terrorists_win	169	26.9
Mirage	NA	12	1.9

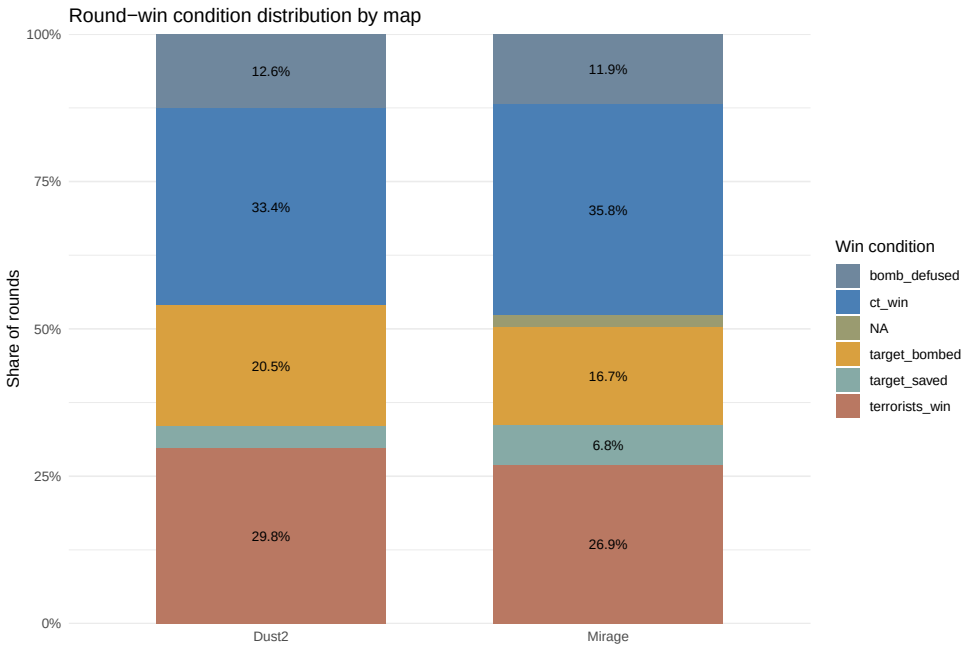


Figure 3: Win type distribution by map

Figure 4 shows the win shares by side and map among different teams. While the shares are more or less even, it can be inferred that some teams are better at playing the T side, likely due to being better attackers, and some being better on the CT side, likely being better defenders.

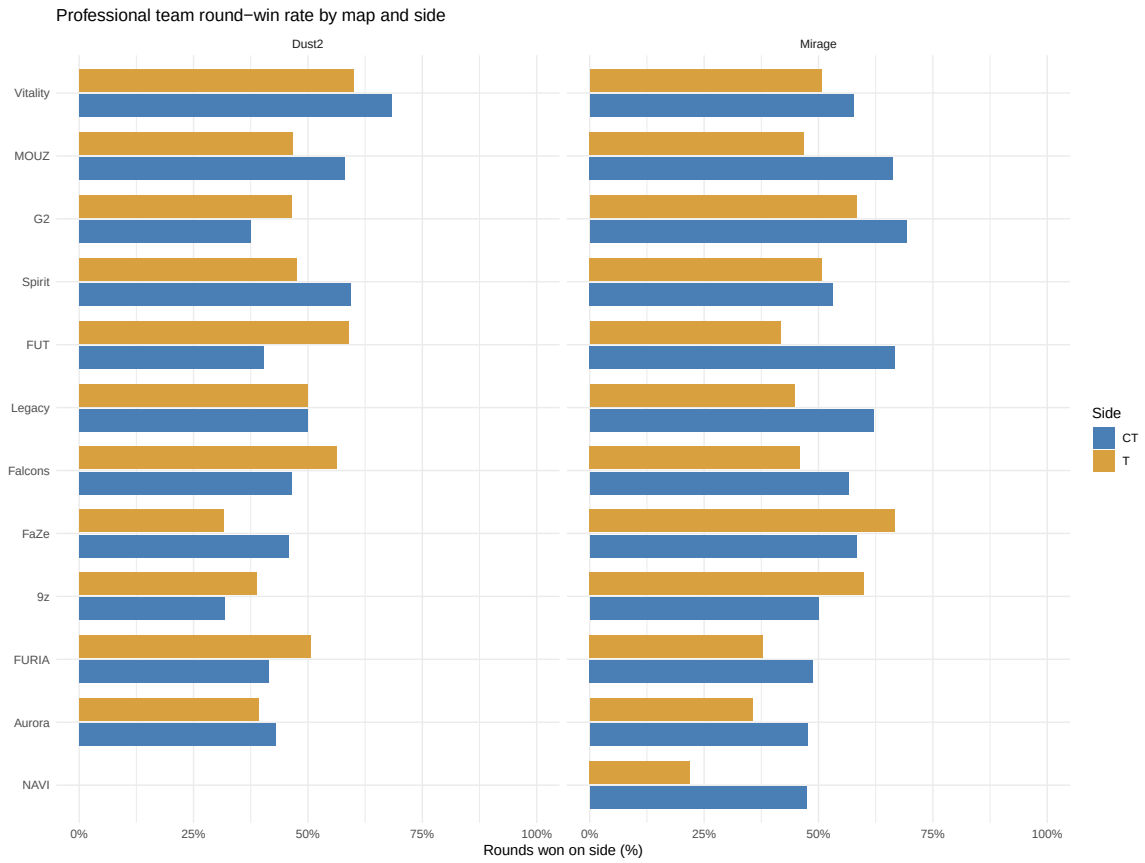


Figure 4: Share of rounds won by side, map and team

Figure 5 shows a heatmap of CT win rate based on how many players are alive on each side. Unsurprisingly, CT win rates remain high if the CT side has an advantage in player number, and vice versa. The seemingly peculiar case of CTs losing some matches with 0 T players being alive is due to the CT side killing all T players, but not having enough time to defuse the planted bomb, resulting in a T victory.

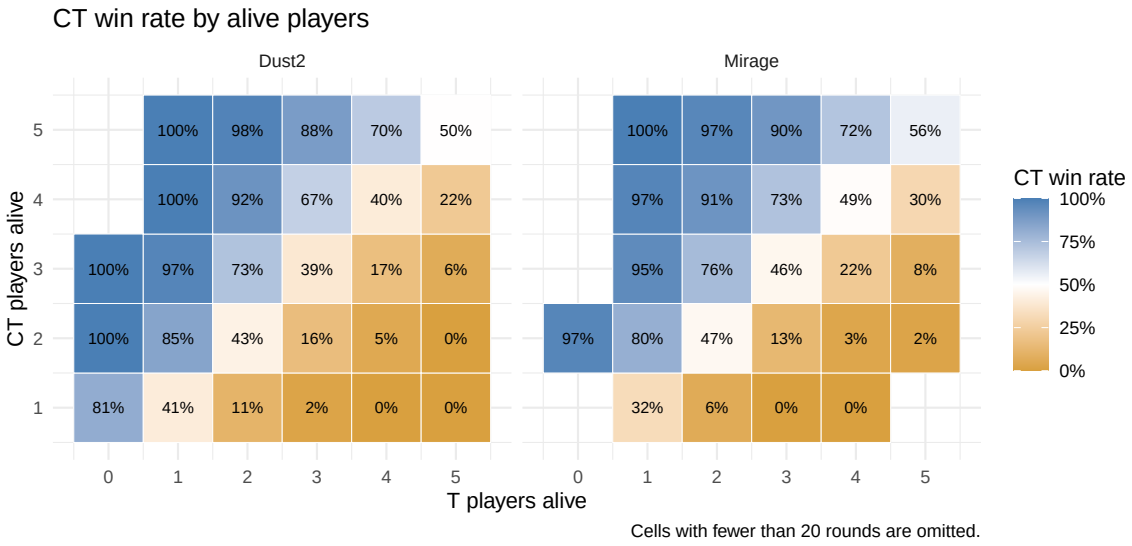


Figure 5: Heatmap of CT win rate according to how many players are alive on each side

Figure 6 shows box plots of the number of players alive at the end of rounds along with their average health. The T side being the attacking side usually has a lower average health at the end of a round. Figure 7 shows more clearly the distribution of alive players by side and map at the end of a round, with the most common occurrence being that only one player survives on both sides.

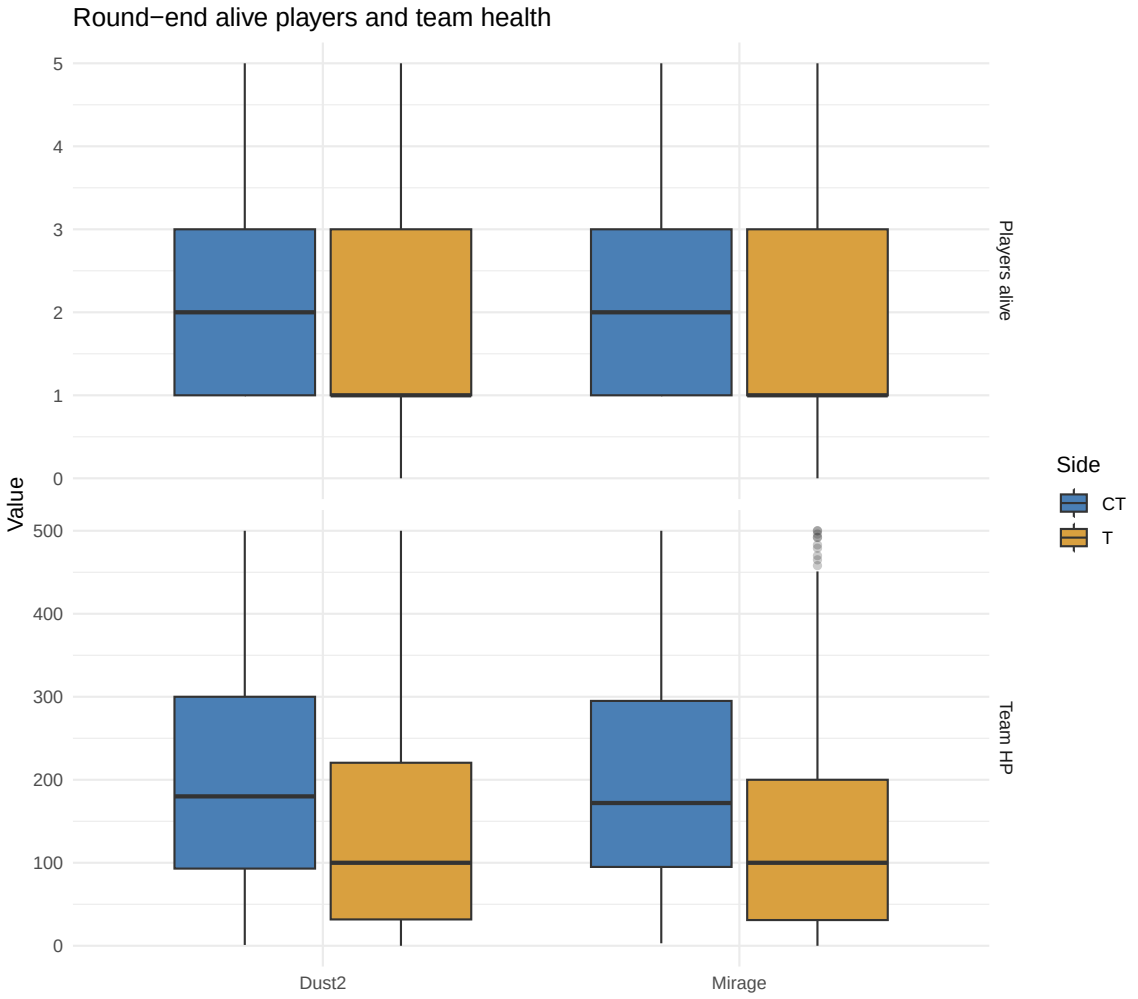


Figure 6: Number of players and their average health at the end of round by side and map

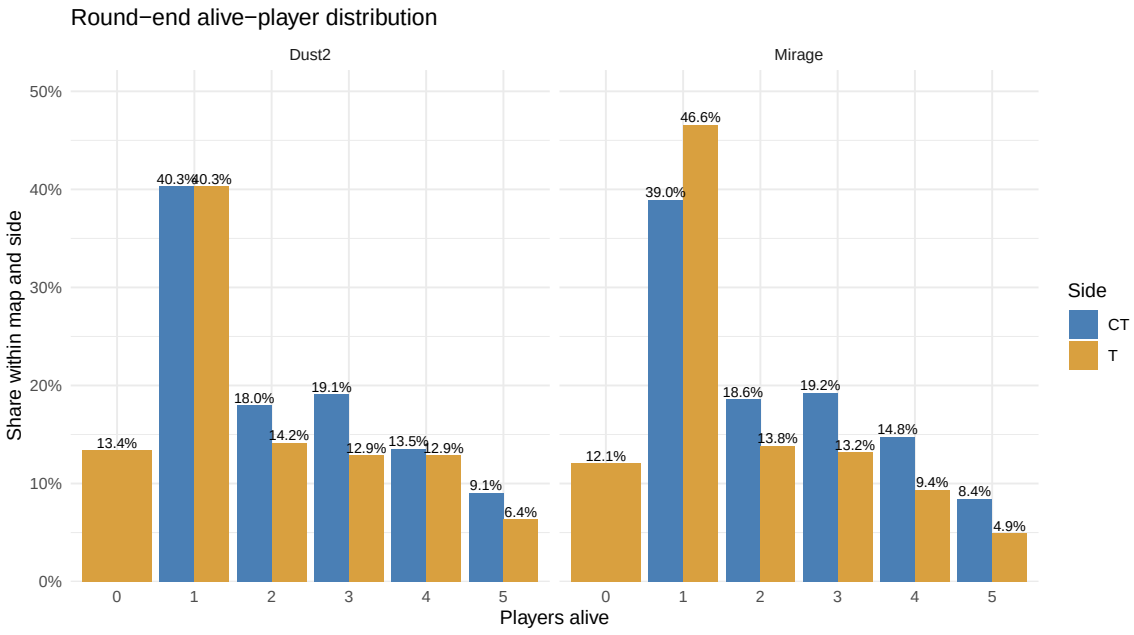


Figure 7: Distribution of alive players at the end of round by side and map

Figure 8 shows the percentage of rounds that had bomb plants occur. As was mentioned previously, Mirage is a map more favourable to the CT side, which could explain the lower number of bomb plants as compared to Dust2, which is a more balanced map for both sides. Figure 9 also shows that in both maps, bomb plants occurred more frequently in the A site rather than B. However, as can be seen in figure 10, the T side would win a larger share of rounds on both maps if the chosen site was B instead of A, perhaps indicating that the B site is easier to defend, thus when T players successfully plant in the B site and start defending, with the CTs attacking, the T side is more likely to win the round.

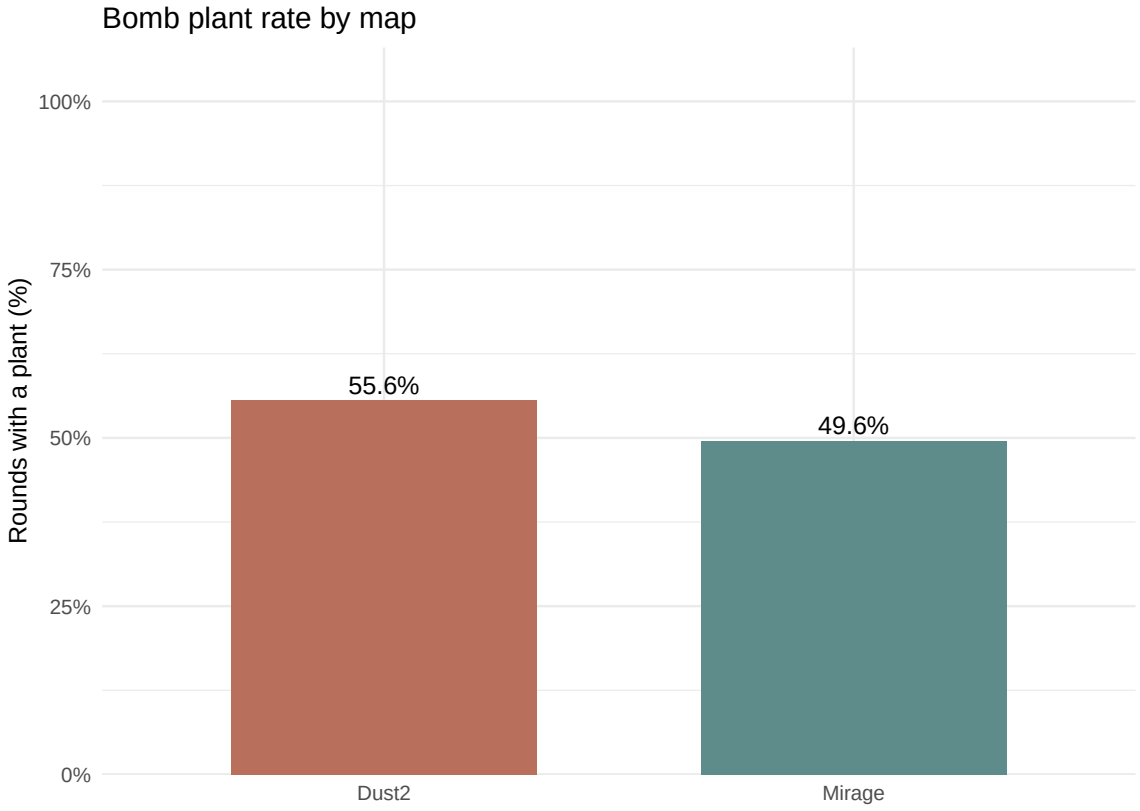


Figure 8: Percentage of rounds with bomb plant by map

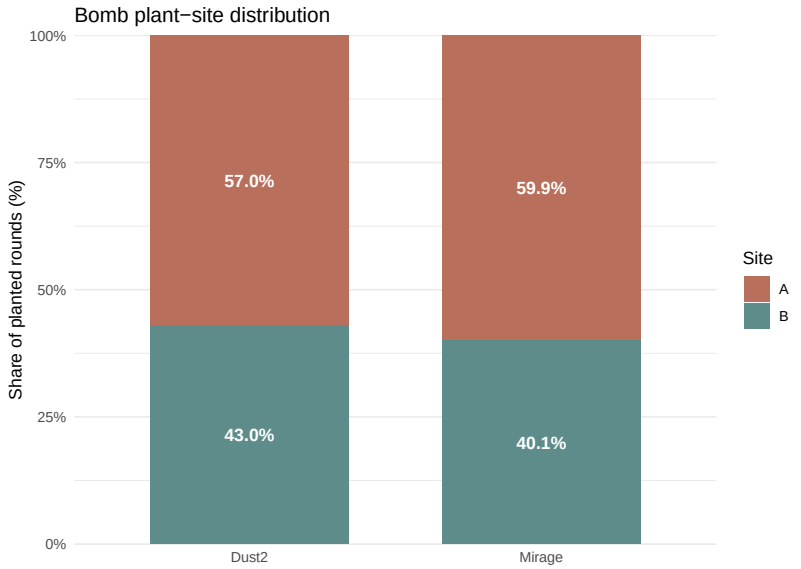


Figure 9: Distribution of which bomb site was chosen for planting by map

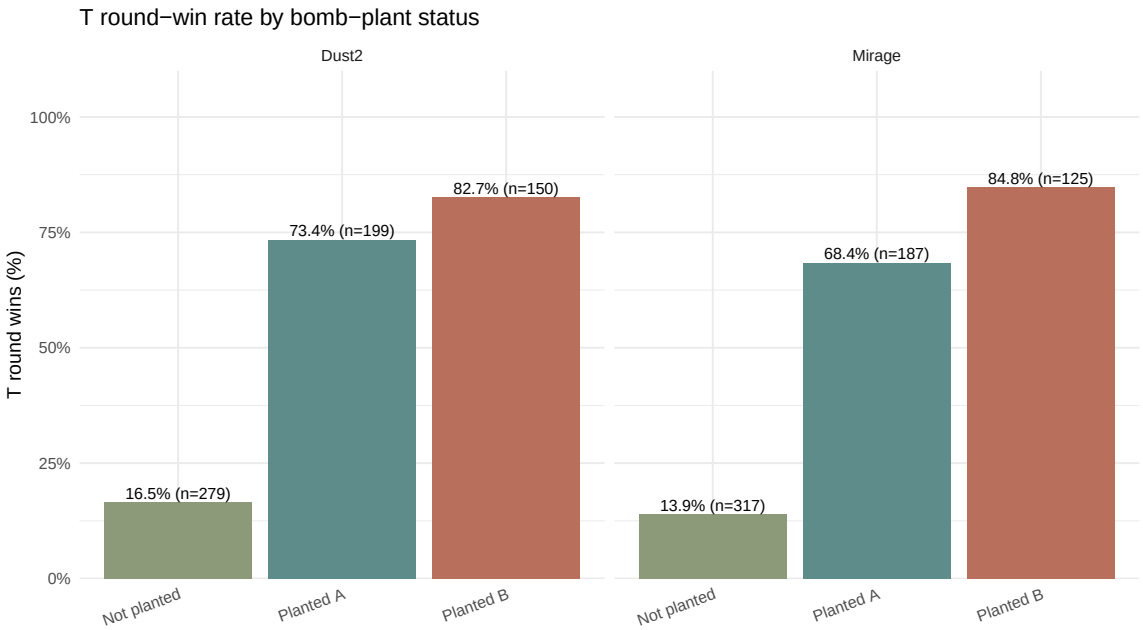


Figure 10: Distribution of T side wins by bomb plant and bomb site choice status

Another interesting area of exploration is how the buy-type affects win rate. A buy type indicates the strategy used to purchase equipment under different economic scenarios. A pistol buy (or pistol round) occurs during the first round of a match (or after switching sides at the half point), where players have only enough money to buy a secondary weapon (a pistol) and/or a utility grenade. A full buy means that players buy a primary weapon, armor and utilities according to their role, and this usually occurs when the players have money to spare. An eco round is the opposite, usually occurring when players have little money, which usually means purchasing a cheaper primary weapon, forgoing utilities or armor, or even purchasing nothing. A force buy occurs when players have little money, and either commit to a full buy leaving them with no money in case of a round loss, or spending what little money they have on the best equipment possible. A force buy is usually done as a risky strategy to break a losing streak, as winning rounds rewards the winning side with more money than losing.

Table 5 shows the CT win rates according to the CT and T buy type, and figure 11 shows a heatmap of these win rates. Pistol rounds and full buy - full buy matchup rounds result in a fairly even CT - T win rate. Interestingly, in a force buy - force buy matchup, the T side is far more likely to win than the CT side.

Table 5: CT and T buy types with corresponding CT win rates

CT buy type	T buy type	% CT win rate	Number of CT win rounds
pistol	pistol	0.53	120
eco	force	0	5
eco	full	0.075	40
force	eco	1	12
force	force	0.22	37
force	full	0.21	150
full	eco	0.92	78
full	force	0.72	191
full	full	0.54	613

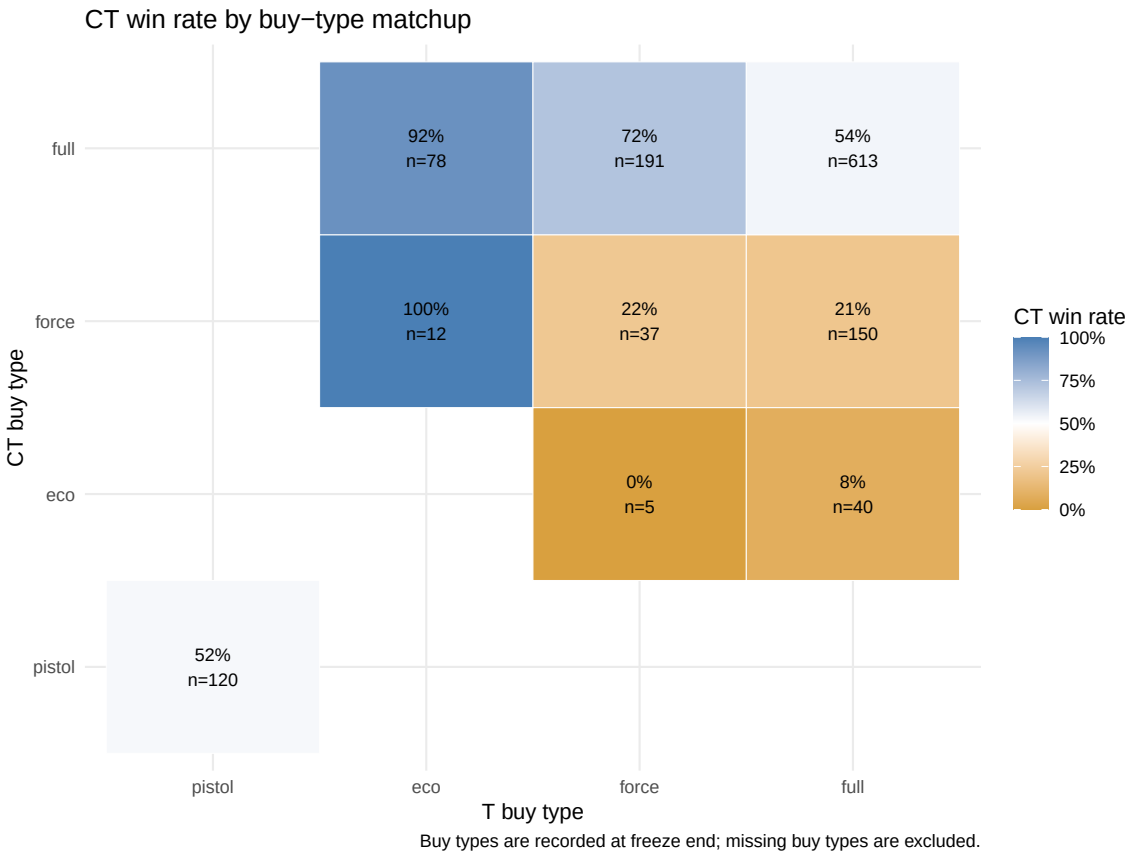


Figure 11: CT Win rate by side buy type across both maps

Data was also explored at the (approximately) 30 second snapshot, to see how CT win rates change with one side having more players alive (Table 6, Figure 12), better equipment (Table 7) or having an advantage in player health point and armor conditions (Table 8). Perhaps unsurprisingly, having an advantage in player count, equipment value, health and armor results in a higher probability of winning the round.

Table 6: Alive player by side advantage and CT win rates with corresponding statistics, at 30s snapshot. Rounds indicates how many rounds had such an advantage.

Advantage	Rounds	% CT win
More CTs alive	300	81
Equal player counts	749	50.7
More Ts alive	208	19.2

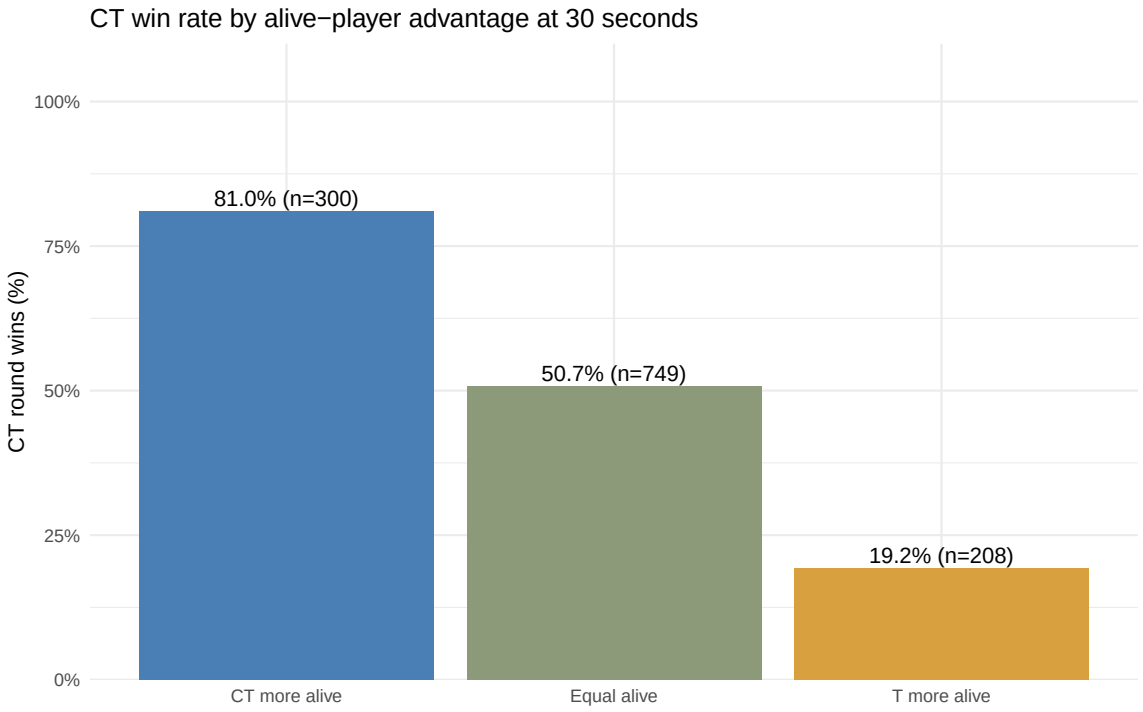


Figure 12: Change in CT win rate by side alive player count advantage

Table 7: Equipment advantage and CT win rates with corresponding statistics, at 30s snapshot. Rounds indicates how many rounds had such an advantage. Means calculated by taking every round (CT - T) value and then taking the mean of those values

Advantage	Rounds	% CT win	Mean equip. val. diff.	Mean money diff.
CT higher equipment value	793	68.7	9770	5265
Equal equipment value	6	50	0	-617
T higher equipment value	458	25.1	-8162	-5151

Table 8: Player health point and armor advantage and CT win rates with corresponding statistics, at 30s snapshot. Means calculated by taking every round (CT - T) value and then taking the mean of those values

Advantage	Rounds	% CT Win	Mean HP diff.	Mean armor diff.
CTs higher HP	551	68.4	95.5	120.1
Equal HP	342	51.8	0	27.6
Ts higher HP	364	29.9	-76.7	-84.4

This concludes the exploratory data analysis. The data and statistics chosen here are only a fraction of the data that is possible to visualize and explore, but nonetheless even in this narrow slice, some noteworthy findings showed up, e.g. the bomb site choice distribution compared to the win rates by bomb site choice.

References

João M. Alves and Ramiro S. Barbosa. Machine learning for basketball Game Outcomes: NBA and WNBA leagues. *Computation*, 13(10):230, 10 2025. doi: 10.3390/computation13100230. URL https://doi.org/10.3390/computation13100230.

Daniel Berrar, Philippe Lopes, and Werner Dubitzky. A data- and knowledge-driven framework for developing machine learning models to predict soccer match outcomes. *Machine Learning*, 113(10):8165–8204, 9 2024. doi: 10.1007/s10994-024-06625-9. URL https://doi.org/10.1007/s10994-024-06625-9.

Arvid Björklund, Fredrik Lindevall, Philip Svensson, and William Johansson Visuri. Predicting the outcome of CS:GO games using machine learning. Bachelor of science thesis, Chalmers University of Technology, 2018. URL https://hdl.handle.net/20.500.12380/256129. Report DATX02-18-11.

Wei-Jen Chen, Mao-Jhen Jhou, Tian-Shyug Lee, and Chi-Jie Lu. Hybrid basketball game outcome prediction model by integrating data mining methods for the National Basketball Association. *Entropy*, 23(4):477, 2021. doi: 10.3390/e23040477.

d-roho. CSGOPredictor. https://github.com/d-roho/CSGOPredictor, n.d. GitHub repository; accessed 2026-10-23.

eigenpaul. cs2-evalbar. https://github.com/eigenpaul/cs2-evalbar, n.d. GitHub repository; accessed 2026-10-23.

Youssof Eshrah and Yomna M. I. Hassan. CS2AnalyticsAPP: A multi-player shooter game analytics framework. In *2025 35th International Conference on Computer Theory and Applications (ICCTA)*, pages 119–124, Alexandria, Egypt, 2025. doi: 10.1109/ICCTA68914.2025.11519898.

Gary2005. cs-net: A transformer-based framework for Counter-Strike 2 demo analysis. https://github.com/Gary2005/cs-net, n.d. GitHub repository; accessed 2026-10-23.

Shuoyu Jing, Mohd Mahzan Awang, Wan Ahmad Munsif Wan Pa, Pengyu Pan, Daniel Memmert, and Zien Zhang. Interpretable machine learning with SHAP for esports performance analysis of professional Counter-Strike players from 2012 to 2025. *International Journal of Sports Science & Coaching*, 21(5):3298–3314, 2026a. doi: 10.1177/17479541251388864. First published online 2025-10-24; issue published 2026-10.

Shuoyu Jing, Mohd Mahzan Awang, Wan Ahmad Munsif Wan Pa, Pengyu Pan, and Zhibin Zhang. In-game behavior and performance: A comparative network analysis of esports performance among professional Counter-Strike players 2012–2025. *International Journal of Human–Computer Interaction*, 2026b. doi: 10.1080/10447318.2026.2666667. Published online 2026-05-13.

Ilya Makarov, Dmitry Savostyanov, Boris Litvyakov, and Dmitry I. Ignatov. Predicting winning team and probabilistic ratings in "Dota 2" and "Counter-Strike: Global Offensive" video games. *Lecture notes in computer science*, pages 183–196, 12 2017. doi: 10.1007/978-3-319-73013-4\{_}17. URL https://doi.org/10.1007/978-3-319-73013-4_17.

Ebenezer Fiifi Emire Atta Mills, Zihui Deng, Zhuoqing Zhong, and Jinger Li. Data-driven prediction of soccer outcomes using enhanced machine and deep learning techniques. *Journal Of Big Data*, 11(1), 11 2024. doi: 10.1186/s40537-024-01008-2. URL https://doi.org/10.1186/s40537-024-01008-2.

Diego Moya, Christian Tipantuña, Génesis Villa, Xavier Calderón-Hinojosa, Belén Rivadeneira, and Robin Álvarez. Machine Learning Applied to Professional Football: performance improvement and results prediction. *Machine Learning and Knowledge Extraction*, 7(3):85, 8 2025. doi: 10.3390/make7030085. URL https://doi.org/10.3390/make7030085.

Yan Ouyang, Xuewei Li, Wenjia Zhou, Wei Hong, Weitao Zheng, Feng Qi, and Liming Peng. Integration of machine learning XGBoost and SHAP models for NBA game outcome prediction and quantitative analysis methodology. *PLOS ONE*, 19(7):e0307478, 2024. doi: 10.1371/journal.pone.0307478.

ratulsarker. CS2_RoundPrediction: CS2 round winner prediction using deep learning. https://github.com/ratulsarker/CS2_RoundPrediction, n.d. GitHub repository; accessed 2026-10-23.

Qiyuan Shen. A machine learning approach to predict the result of League of Legends. In *2022 International Conference on Machine Learning and Knowledge Engineering (MLKE)*, pages 38–45, 2022. doi: 10.1109/MLKE55170.2022.00013.

Stanlly, Fauzan Ardhana Putra, and Nunung Nurul Qomariyah. DOTA 2 win loss prediction from item and hero data with machine learning. In *2022 IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communications Technology (IAICT)*, pages 204–209, 2022. doi: 10.1109/IAICT55358.2022.9887525.

TaiZo1. cs2-win-prediction. `https://github.com/TaiZo1/cs2-win-prediction`, n.d. GitHub repository; accessed 2026-10-23.

wyn0n0. CSMapWinPredict: Round-level win-probability modeling and replay analytics for CS2. `https://github.com/wyn0n0/CSMapWinPredict`, n.d. GitHub repository; accessed 2026-10-23.

Xinyao Xia, Abel Salinas, and Fred Morstatter. Precision under fire: Analysis and predictive modeling in Counter-Strike: Global offensive. In *2025 IEEE Conference on Games (CoG)*, pages 1–8, 2025. doi: 10.1109/CoG64752.2025.11114164.

Appendix A. Variables in the rounds and rounds_states datasets along with their data types.

Table 9: Variables in the rounds and rounds_states datasets along with their data types.

Beginning of Table 9		
Variable	Dataset	Variable data type
bomb_planted	Both rounds and round_states	logical
ct_score_before	Both rounds and round_states	integer
ct_team	Both rounds and round_states	character
ct_win	Both rounds and round_states	logical
map	Both rounds and round_states	character
match_id	Both rounds and round_states	character
part	Both rounds and round_states	integer
plant_site	Both rounds and round_states	character
round_num	Both rounds and round_states	integer
t_score_before	Both rounds and round_states	integer
t_team	Both rounds and round_states	character
winner_side	Both rounds and round_states	character
bomb_carried	round_states	logical
ct_alive	round_states	integer
ct_armor	round_states	integer
ct_awps	round_states	integer

Continuation of Table 9		
Variable	Dataset	Variable data type
ct_defusers	round_states	integer
ct_equipment	round_states	integer
ct_flashes	round_states	integer
ct_he	round_states	integer
ct_helmets	round_states	integer
ct_hp	round_states	integer
ct_molotovs	round_states	integer
ct_money	round_states	integer
ct_rifles	round_states	integer
ct_smgs	round_states	integer
ct_smokes	round_states	integer
seconds	round_states	numeric
seconds_since_plant	round_states	numeric
t_alive	round_states	integer
t_armor	round_states	integer
t_awps	round_states	integer
t_defusers	round_states	integer
t_equipment	round_states	integer
t_flashes	round_states	integer
t_he	round_states	integer
t_helmets	round_states	integer
t_hp	round_states	integer
t_molotovs	round_states	integer
t_money	round_states	integer
t_rifles	round_states	integer
t_smgs	round_states	integer
t_smokes	round_states	integer
tick	round_states	integer
time_remaining_s	round_states	numeric
ct_buy_type	rounds	character
ct_equipment_freeze_end	rounds	integer
ct_won_previous_round	rounds	logical
end_tick	rounds	integer
first_death_side	rounds	character
freeze_end_tick	rounds	integer
is_overtime	rounds	logical
is_pistol_round	rounds	logical
plant_tick	rounds	integer
reason_name	rounds	character
round_duration_s	rounds	numeric
t_buy_type	rounds	character

Continuation of Table 9		
Variable	Dataset	Variable data type
t_equipment_freeze_end	rounds	integer
winner_team	rounds	character